

Back Jer as a Negative Marker: Jer Color in Slavic Liquid Contexts as an Etymological Diagnostic and Evidence for Syllabic-Resonant Coloring

Marko Jevremović

Independent researcher, Belgrade, Serbia

`marko@asemic.me`

May 2026

Preprint — not yet peer-reviewed

Abstract

Old Church Slavonic distinguishes two jer letters, ѣ (back) and ѥ (front), whose distribution in liquid contexts (*CъrC/CьrC*, *CъlC/CьlC*) has resisted phonological explanation for over a century. This paper makes a coordinated empirical and theoretical claim. Empirically, on the basis of a 128-word Proto-Slavic corpus together with a reanalysis of Kurylowicz’s (1956) survey of Trautmann’s (1924) Balto-Slavic dictionary, jer color in these positions is shown to function as an etymological diagnostic: front jer ѥ marks inheritance from a PIE syllabic resonant (**r*, **l*); back jer ѣ marks every other source—loanwords, laryngeal sequences, substrate vocabulary, full-grade roots. The diagnostic is categorical for unconditioned inherited vocabulary in both directions: among confirmed inherited front-jer-plus-liquid words ($n = 60$), none fails to trace to a PIE syllabic resonant; among back-jer-plus-liquid words with no labial/round conditioning environment ($n = 50$), none traces to a PIE syllabic resonant. Seven cases of PIE syllabic resonants yielding back jer (**gъrdlo*, **gъrnъ*, **bъrzъ*, **drъvo*, **tъrkъ*, **mъlva*, **tъlkъ*) are explained by a unified labial/round conditioning environment (labiovelars, *u*-stems, labial **w*). Theoretically, the conditioning is uniformly consonantal and glide-based, with no vowel-quality features as conditioners; this argues that the segment being conditioned was a colored syllabic consonant rather than a reduced vowel, and that OCS ѣ/ѥ in liquid contexts function as notational conventions for that syllabic resonant. The argument converges with the Havlík’s-law vocalization asymmetry developed in a companion preprint (Jevremović 2026) and reframes the etymological diagnostic as a downstream consequence of a structural account in which inherited PIE material and loaned or substratum material feed the same Slavic syllabic-resonant slot through two distinct channels.

Keywords: Proto-Slavic, jers, syllabic resonants, liquid contexts, etymological classification, Old Church Slavonic phonology, Winter’s law

1 Introduction

The distribution of back jer ɤ and front jer ɛ in Proto-Slavic (PS) words where a jer is adjacent to a liquid (*r* or *l*) has long been recognized as etymologically conditioned, but the full implications of this conditioning have not been drawn out. The principal published corpus is Kuryłowicz’s (1956: 227–242) survey of all liquid- and nasal-context items in Trautmann’s (1924) *Baltisch-Slavisches Wörterbuch*, an exhaustive Balto-Slavic etymological dictionary. Kuryłowicz tabulated 258 etymologies, of which 200 show only **iR* and 58 show **uR*, broken down by preceding-consonant environment (Table 1). Andersen (2003: 60) recounted the corpus into 157 (73%) pure **iR*, 36 (17%) pure **uR*, and 22 (10%) variable items (giving 215, somewhat fewer than Kuryłowicz; the difference reflects different treatment of Slavic–Baltic doublets).

Table 1: Kuryłowicz’s (1956: 229) breakdown of PBS **iR*/**uR* reflexes by preceding-segment environment.

Environment	<i>*iR</i>	<i>*uR</i>	% <i>*uR</i>
After palatovelars (<i>*k̑</i> , <i>*g̑</i>)	16	0	0%
Before <i>u</i> or laryngeal	35	3	7.9%
Initial	10	1	9%
After dentals (<i>*t</i> , <i>*d</i>)	31	7	18.4%
After labials (<i>*p</i> , <i>*b</i>)	24	6	20%
After liquids (<i>*l</i> , <i>*r</i>)	22	6	21.4%
Before <i>i</i>	2	1	33.3%
After <i>*m</i> , <i>*n</i>	15	8	34.8%
After velars (<i>*k</i> , <i>*g</i>)	38	21	35.6%
After <i>*s</i>	5	4	44.4%
Total	200	58	22.5%

Two features of Kuryłowicz’s breakdown are critical. First, after palatovelars (**k̑*, **g̑*), **uR* appears in **zero of 16** cases—i.e., Kuryłowicz himself established categorically that palatovelars never produce **uR*. Second, the highest **uR* rates appear after velars (35.6%, which Kuryłowicz does not subdivide into plain velars and labiovelars), after **s* (44.4%), and after **m*, **n* (34.8%); after liquids and labials the rates fall to 20% range; before laryngeals or in initial position they collapse to under 10%. The pattern is consistent with the picture developed in §3: rounded segments (labiovelars within the velar count; labial **w* adjacent to a labial; *u*-stem suffixes) raise the **uR* rate, while environments without a labial/round element retain the default **iR*.

Multiple partial conditioning environments have been proposed in the literature: Vaillant (1950) observed that labiovelars condition **uR*; Kuryłowicz (1956: 229) noted (with Vaillant) that “*ur* is reliable in Slavic only after a guttural”; Shevelov (1964: 86–91), reviewing the Slavic data in detail, concluded that the distribution is not reducible to phonological conditioning; Stang (1966) suggested that **iR* is the productive gram-

matal reflex (zero-grades, ablaut alternations) while $*uR$ clusters with expressive and onomatopoetic vocabulary; Matasović (2008: 111) proposed an ordered-rule derivation in which the prothetic schwa underlying $**_R$ goes to $**_u$ after velars before nasals and to $**_i$ elsewhere. None of these proposals individually eliminates the residue, and Andersen (2003) leaves the broader distribution open as “an unsolved problem.”

The present paper makes a stronger claim, restricted to Slavic and limited to liquid contexts: a categorical distribution for the unconditioned cleaned corpus (60/60 inherited front-jer items, 50/50 unconditioned back-jer items), where Kuryłowicz reports the broader Balto-Slavic 22.5% $*uR$ rate. The residual gap is hypothesized to reduce under two factors that an aggregate Balto-Slavic count does not separate. First, the $*uR$ cases coincide largely with the conditioning environments isolated in §3 (labiovelars subsumed within Kuryłowicz’s “after velars” category, u -stems, labial $*w$), generalizing Vaillant. Second, the “variable” Slavic/Baltic doublets consist largely of items where one branch underwent the labial/round conditioning while the other did not, and are silent about Slavic’s internal distribution. Once the conditioning is held constant and the question is restricted to Slavic, the residue is predicted to collapse. A complete item-by-item reanalysis of Kuryłowicz’s 58 $*uR$ examples (1956: 229–234), restricted to Slavic-internal liquid contexts, is undertaken in §6.4 below.

In a companion paper (Jevremović 2026), we documented a categorical asymmetry in Havlík’s law outcomes: Czech strong-position jers vocalize to /e/ in 100% of non-liquid cases but 0% of liquid-adjacent cases, suggesting that liquid-context jers were not real reduced vowels but notational conventions for syllabic consonants inherited from PIE. The present paper pursues a different implication of the same observation: if jer color in liquid contexts preserves etymological information from PIE, it can be used as a **diagnostic tool** for classifying Proto-Slavic vocabulary.

The claim is simple:

- **Front jer ь + liquid** $\Rightarrow$ the word inherits a PIE syllabic resonant ($*_r$ or $*_l$).
- **Back jer ɤ + liquid** $\Rightarrow$ the word does **not** inherit a PIE syllabic resonant. Its source is a loanword, a laryngeal sequence, a substrate word, or a full-grade/ o -grade root.

The second half of this claim—the negative diagnostic—is categorical for unconditioned items: in a corpus of ~ 128 words, no Proto-Slavic word with back jer adjacent to a liquid derives from a PIE syllabic resonant unless a labial/round conditioning environment (§3) is present. The first half is categorical for inherited vocabulary ($n = 60$, all tracing to PIE syllabic resonants), with the only front-jer words that do *not* trace to syllabic resonants being four late Christian-era loanwords adapted to a productive phonotactic template. The categorical-for-cleaned-corpus framing should be read with

the carve-outs in mind: seven conditioned cases are explained by an explicit closed-list mechanism (§3), and fifteen items fall in an “uncertain/substrate/expressive” subgroup whose treatment is documented in Table 4.

The paper develops the diagnostic empirically first (§2–§3) and then, in a series of theoretical subsections (§3.5–§3.7), argues that the diagnostic is not a free-standing empirical correlation but a downstream consequence of a structural account: the OCS jer letter in liquid contexts notates a colored syllabic consonant, and the etymological correlation arises from how PIE inheritance, loanwords, and substrate material are mapped onto the same syllabic-resonant slot.

The paper is organized as follows. Section 2 presents the corpus and the core 2×2 contingency table. Section 3 discusses the labial/round conditioning environments that produce apparent exceptions. Section 4 applies the diagnostic to disputed etymologies. Section 5 compares the jer diagnostic with Winter’s law. Section 6 addresses limitations, and Section 7 concludes.

2 The corpus

2.1 Methodology

The corpus was assembled from three primary sources: (1) Wiktionary’s Proto-Slavic reconstruction namespace (used as a structured index of headwords), (2) Derksen’s *Etymological Dictionary of the Slavic Inherited Lexicon* (2008), and (3) Vasmer’s *Etymological Dictionary of the Russian Language* (online edition). Headwords with disputed or contested reconstructions were cross-checked against Sławski’s *Słownik prasłowiański* and the Brno *Etymologický slovník jazyka staroslověnského* (ESJS); the larger *Etimologičeskij slovar’ slavjanskix jazykov* (ÈSSJa, Trubačev et al.) was consulted in spot checks where Wiktionary and Derksen diverged. Every Proto-Slavic word containing a jer adjacent to a liquid (*r* or *l*)—i.e., matching the patterns *C̣̣̣rC*, *C̣̣̣lC*, *C̣̣̣rC*, *C̣̣̣lC*, or their morpheme-boundary variants—was collected exhaustively within the headword set of the three primary sources, not selected by curation. Each word was classified by its PIE etymological source as either (a) derived from a PIE syllabic resonant (**r̥* or **l̥*), or (b) derived from another source. Modern reflexes in two to four Slavic languages were recorded for verification.

Two front-jer entries (**ṿ̣̣rba* ‘willow’ and **ḅ̣̣rdo* ‘hill’) are not in Derksen (2008) but are included here on the basis of independent comparative evidence: Baltic cognates with **i* vocalism (Lith. *virbas*, Latv. *birds*), Greek cognate *ὄρπηξ* ‘branch’ for **ṿ̣̣rba*, and—for **ḅ̣̣rdo*—an acute accent on Serbo-Croatian *ḅ̣̣rdo* of the form predicted by Hirt’s law (the retraction of accent onto a non-final syllable preceding a laryngeal, modern formulation in Illič-Svityč 1963), which provides independent evidence for the underlying zero-grade

***b^hrh₃-d^ho-* (§4.1, §4.2). Their inclusion is flagged in Table 3; removing them does not alter the diagnostic’s claim for the remaining 58 entries.

The corpus comprises 128 words: 71 with front jer Ъ (60 confirmed inherited, 7 uncertain, 4 loanwords) and 57 with back jer Ь.

2.2 Minimal pairs

The diagnostic is crystallized by two minimal pairs, one nominal and one verbal:

Table 2: Two minimal pairs distinguishing inherited PIE syllabic resonants (front jer) from other sources (back jer).

PS form	Meaning	Jer	PIE source	Origin
<i>*dǝlgǝ</i>	‘long’	Ъ (front)	<i>*dl̥h₁g^hos</i>	Inherited (syllabic <i>*l̥</i>)
<i>*dǝlgǝ</i>	‘debt’	Ь (back)	Gothic <i>dulgs</i>	Loanword (Germanic)
<i>*tǝrzati</i>	‘to tear’	Ъ (front)	<i>*tr̥g^h-</i>	Inherited (syllabic <i>*r̥</i>)
<i>*tǝrgati</i>	‘to tear’	Ь (back)	No regular PIE source	Substrate-derived

The first pair shares the consonant skeleton *d-l-g* and differs only in jer color. The front jer in **dǝlgǝ* ‘long’ marks inheritance from PIE **dl̥h₁g^hos*, with syllabic **l̥*. The back jer in **dǝlgǝ* ‘debt’ marks a Gothic loanword, where Germanic **u* was adapted as PS Ь. Modern reflexes preserve the semantic distinction: Russian долгий ‘long’ vs. долг ‘debt’; Czech *dlouhý* ‘long’ vs. *dluh* ‘debt’. The two channels through which these contrasting inputs feed the same Slavic syllabic-resonant slot are formalized in §3.7: **dǝlgǝ* comes through the inherited-PIE channel with default front coloring, **dǝlgǝ* through the source-vocalism channel with the Gothic **u* preserved as back coloring.

The second pair extends the pattern into verbal semantics. **tǝrzati* (OCS *трѣзаму*, Russian *терзать*, Bulgarian *мѣрзам*, SC *trzati*) derives regularly from PIE **tr̥g^h-* (zero-grade of **terg^h-* ‘to rub’), with Slavic /z/ from the satem reflex of palatovelar **g^h* and front jer from the syllabic resonant. **tǝrgati* (OCS *трѣгѣхму*, Russian *расторгать*, Czech *trhat*, Polish *targać*) has no regular PIE source for its back jer: the three competing proposals in the etymological literature (Chernykh’s **terg^h-*, Derksen’s link to Latin *tergō*, Vasmer’s link to Sanskrit *str̥hāti*) all involve zero-grades that should yield *front* jer (§2.3). The simpler reading is that **tǝrgati* is non-inherited, paired with the substrate noun **tǝrgǝ* ‘market’ (Table 4). The Sanskrit comparison is treated as a worked example in §4.5.

2.3 The front-jer corpus

Table 3 lists all 60 confirmed front-jer words with PIE syllabic resonant etymologies. The corpus also contains 7 uncertain words (with debated etymologies) and 4 late Germanic

loanwords adapted to the productive $\text{ɸ} + \text{liquid}$ template (see below). Derivatives are marked with $\leftarrow$.

Table 3: Front-jer (ɸ) + liquid words from PIE syllabic resonants (complete confirmed corpus, $n = 60$).

#	PS form	Meaning	PIE source	Notes
<i>PIE syllabic *l</i> ($n = 13$)				
1	* <i>ɸɪlk̑</i>	‘wolf’	* <i>wl̑k̑^wos</i>	
2	* <i>ɸɪln̑</i>	‘full’	* <i>pl̑h₁nos</i>	
3	* <i>ɸɪlna</i>	‘wool’	* <i>h₂wl̑h₁neh₂</i>	
4	* <i>žɪlt̑</i>	‘yellow’	* <i>ǵ^hl̑h₃tós</i>	
5	* <i>dɪlg̑</i>	‘long’	* <i>dl̑h₁ǵ^hos</i>	Minimal pair
6	* <i>žɪlv̑</i>	‘turtle’	* <i>ǵ^hl̑h₃w-</i>	Cf. #4
7	* <i>tɪlkn̑ti</i>	‘to push’	* <i>tl̑k-</i>	Baltic * <i>i</i>
8	* <i>dɪlbati</i>	‘to hollow’	* <i>d^hl̑b^h-</i>	Cf. OE <i>delfan</i>
9	* <i>tɪlo</i>	‘floor’	* <i>tl̑h₂-o-</i>	
10	* <i>bɪltati</i>	‘to babble’	* <i>b^hl̑-t-</i>	Cf. Lith. <i>bilsti</i>
11	* <i>m̑l̑cati</i>	‘to be silent’	* <i>ml̑k-?</i>	PBS has * <i>i</i>
12	* <i>tɪl̑ti</i>	‘to rot’	* <i>tl̑H-?</i>	Cf. #9
13	* <i>čɪln̑</i>	‘boat’	* <i>(s)kl̑-no-s</i>	Lith. <i>kėlnas</i>
<i>PIE syllabic *r</i> ($n = 39$)				
14	* <i>m̑ɪrtv̑</i>	‘dead’	* <i>m̑rtwós</i>	
15	* <i>ɸɪrx̑</i>	‘top’	* <i>w̑rs-</i>	
16	* <i>ɸɪrst̑</i>	‘finger’	* <i>pȓsth₂-</i>	
17	* <i>tɪrn̑</i>	‘thorn’	* <i>tȓnós</i>	
18	* <i>ɸɪrv̑</i>	‘first’	* <i>pȓh₂wós</i>	
19	* <i>z̑rno</i>	‘grain’	* <i>ǵ̑rh₂nóm</i>	
20	* <i>s̑ɪrd̑ce</i>	‘heart’	* <i>k̑rd-</i>	
21	* <i>č̑rv̑</i>	‘worm’	* <i>k̑^w̑rmis</i>	Palatalized labiovelar
22	* <i>č̑rn̑</i>	‘black’	* <i>k̑rsnós</i>	
23	* <i>b̑ɪrv̑no</i>	‘beam’	* <i>b^h̑rw-</i>	
24	* <i>č̑rta</i>	‘line’	* <i>(s)k̑rt-</i>	
25	* <i>s̑ɪrst̑</i>	‘fur’	* <i>s̑rs-ti-</i>	
26	* <i>ɸɪrt̑ti</i>	‘to turn’	* <i>w̑rt-</i>	
27	* <i>d̑ɪȓžati</i>	‘to hold’	* <i>d^h̑ȓǵ^h-</i>	
28	* <i>t̑ɪrd̑</i>	‘hard’	* <i>tw̑rd-</i>	
29	* <i>ž̑ɪrti</i>	‘to sacrifice’	* <i>ǵ^w̑ȓH-</i>	Palatalized labiovelar
30	* <i>t̑ɪrp̑ti</i>	‘to endure’	* <i>t̑rp-</i>	Cf. Latin <i>torpēre</i>

#	PS form	Meaning	PIE source	Notes
31	*s̥br̥na	‘roe deer’	*k̥r̥h₂-	Cf. Latin <i>cervus</i>
32	*s̥br̥p̥θ	‘sickle’	*s̥r̥p̥-	Finnish <i>sirppi</i>
33	*č̥et̥v̥rt̥θ	‘fourth’	*k̥ʷet̥w̥rt̥ós	
34	*b̥br̥ati	‘to take’	*b̥ʰr̥-	
35	*d̥br̥ati	‘to tear’	*d̥r̥H-	
36	*gr̥m̥ēti	‘to thunder’	*g̥ʰr̥m̥-	
37	*m̥br̥zn̥q̥ti	‘to freeze’	*m̥r̥s-	
38	*v̥br̥ba	‘willow’	*w̥r̥b̥ʰ-eh₂	Not in Derksen 2008; §4.1
39	*b̥br̥do	‘hill’	*b̥ʰr̥h₃d̥ʰo-	Not in Derksen 2008; §4.2
40	*v̥br̥ša	‘fish trap’	*w̥r̥ǵ-	Cf. Lith. <i>varžas</i>
41	*v̥br̥ēti	‘to boil’	*w̥r̥H-	
42	*m̥br̥kn̥q̥ti	‘to grow dark’	*m̥r̥Hk-	
43	*ž̥br̥nov̥θ	‘millstone’	*g̥ʷr̥h₂nuH-	Cf. Eng. <i>quern</i>
44	*ž̥br̥d̥b̥	‘pole, rod’	*g̥ʰr̥d̥ʰ-	Probable zero-grade
45	*st̥br̥v̥b̥	‘carrion’	*st̥r̥w-	Probable zero-grade
46	*m̥br̥va	‘crumb’	*m̥r̥w-	Probable zero-grade
47	*m̥br̥cati	‘to grow dark’	*m̥r̥g̥ʷ-	Probable zero-grade
48	*č̥br̥stv̥θ	‘stale’	*k̥r̥st̥-?	Front-er pattern
49	*s̥m̥br̥t̥b̥	‘death’	*m̥r̥tis	ʱ adjacent to <i>r</i>
50	*s̥br̥b̥ati	‘to slurp’	*s̥r̥b̥ʰ-	Zero-grade of *s̥erb̥ʰ-
51	*s̥br̥b̥θ	‘Serb’	*s̥r̥b̥ʰ-os	Schuster-Šewc 1985; §6.5
52	*t̥br̥z̥ati	‘to tear, torment’	*tr̥ǵʰ-	Minimal pair; §2.2
<i>Derivatives (n = 8)</i>				
53	*č̥br̥ven̥θ	‘red’	← *č̥br̥v̥b̥	
54	*ž̥br̥t̥va	‘sacrifice’	← *ž̥br̥t̥i	
55	*ž̥br̥v̥c̥b̥	‘priest’	← *ž̥br̥t̥i	Agent noun
56	*m̥br̥v̥iti	‘to crumble’	← *m̥br̥va	
57	*d̥br̥g̥ati	‘to scrub’	← *d̥br̥ati	
58	*d̥br̥k̥ati	‘to jingle’	← *d̥br̥ati	
59	*d̥br̥p̥ati	‘to pluck’	← *d̥br̥ati	
60	*m̥b̥l̥kn̥q̥ti	‘to fall silent’	← *m̥b̥l̥č̥ati	Inchoative

All 60 confirmed front-er words with established etymologies trace to PIE syllabic resonants. No inherited front-er-plus-liquid word has a non-syllabic-resonant source. The 7 uncertain front-er words (*s̥l̥za ‘tear’, *xr̥bb̥t̥θ ‘spine’, *č̥br̥pati ‘to scoop’, *sm̥br̥d̥ēti ‘to stink’, *p̥l̥z̥ati ‘to crawl’, *č̥br̥t̥θ ‘devil’, *b̥br̥log̥θ ‘den’) have debated PIE etymologies; most are likely syllabic-resonant words whose precise reconstruction

is disputed.

The four front-jer loanwords—**lv̥v̥* ‘lion’ (← Germanic ← Latin *leō*), **kr̥v̥st̥* ‘cross’ (← Germanic ← Latin *Christus*), **c̥rky* ‘church’ (← West Germanic ← Greek *kyriakón*), and **lv̥st̥* ‘ruse’ (← Gothic)—are all Christian-era borrowings that were adapted to the existing productive *v̥+liquid* template. Their existence actually supports the diagnostic: by the late PS period, front jer plus liquid was a productive phonotactic pattern that attracted new words regardless of etymology.

2.4 The back-jer corpus

Table 4 lists the back-jer words, classified by etymological source.

Table 4: Back-jer (ɸ) + liquid words by etymological source
(complete corpus, $n = 57$).

#	PS form	Meaning	Source	Classification
<i>Loanwords (n = 7)</i>				
1	<i>*d̥ɔlg̥</i>	‘debt’	Gothic <i>dulgs</i>	Germanic
2	<i>*x̥ɔlm̥</i>	‘hill’	PGmc <i>*hulmaz</i>	Germanic
3	<i>*p̥ɔlk̥</i>	‘regiment’	PGmc <i>*fulkq</i>	Germanic
4	<i>*x̥ɔrvat̥</i>	‘Croat’	Iranian <i>*xurvāt-</i>	Iranian
5	<i>*k̥ɔlbasa</i>	‘sausage’	Turkic <i>*kul+bas-</i>	Turkic
6	<i>*b̥ɔlv̥an̥</i>	‘idol’	Old Turkic <i>balbal</i>	Turkic
7	<i>*k̥ɔləj̥</i>	‘glue’	cf. MLG <i>kalan</i>	Germanic loan (Kiparsky 1934)
<i>Laryngeal sequences (n = 4)</i>				
8	<i>*k̥ɔrmiti</i>	‘to feed’	<i>*k̥orh̥₃-mo-</i>	Full-grade + laryngeal
9	<i>*g̥ɔrd̥</i>	‘proud’	<i>*g̥^wrH-d̥^hh̥₁o-</i>	Labiovel.+laryng.
10	<i>*r̥ɔdr̥</i>	‘ruddy’	<i>*h̥₁rud̥^h-r̥ɔs</i>	Initial laryngeal
11	<i>*r̥ɔd̥ēti</i>	‘to blush’	<i>*h̥₁rud̥^hēh̥₁ti</i>	Laryngeal
<i>Inherited non-syllabic-resonant (n = 24)</i>				
12	<i>*k̥ɔrk̥</i>	‘bark’	<i>*(s)ker-</i>	Full-grade
13	<i>*pl̥ɔt̥</i>	‘flesh’	<i>*pleth̥₂-</i>	Full-grade
14	<i>*st̥ɔlb̥</i>	‘pillar’	cf. Lith. <i>stul̥bas</i>	Full-grade
15	<i>*z̥ɔl̥</i>	‘evil’	<i>*g̥^hwel-</i>	Non-syllabic
16	<i>*z̥ɔlo</i>	‘evil (n.)’	← <i>*z̥ɔl̥</i>	Derivative
17	<i>*k̥ɔl̥</i>	‘stake’	zero-grade of <i>*k̥olti</i>	Zero-grade <i>*o</i>
18	<i>*kr̥ɔt̥</i>	‘mole’	<i>*(s)ker-</i>	Full-grade
19	<i>*p̥ɔrt̥</i>	‘rag’	<i>*per-</i>	Full-grade
20	<i>*k̥ɔrpati</i>	‘to patch’	PBS <i>*k̥úr̥pā</i>	PBS <i>*u</i>
21	<i>*s̥ɔln̥ɔce</i>	‘sun’	<i>*s̥óh̥₂w̥l̥</i>	<i>*u</i> -coloring

#	PS form	Meaning	Source	Classification
22	* <i>gōlčēti</i>	‘to speak’	* <i>gol-</i>	Full-grade
23	* <i>bōrščv</i>	‘hogweed’	* <i>b^hres-t-yos</i>	Not syllabic * <i>ɣ</i>
24	* <i>tōrčati</i>	‘to run’	← * <i>tōrkō</i>	Derived from * <i>tōrkō</i>
25	* <i>kōlzōkō</i>	‘slippery’	← * <i>kōlzati</i>	Derived from * <i>kōlzati</i>
26	* <i>kōrčiti</i>	‘to cramp’	← * <i>kōrčv</i>	Derived from * <i>kōrčv</i>
27	* <i>bōrna</i>	‘lip, snout’	cf. Arm. <i>beran</i>	Uncertain
28	* <i>tōlpa</i>	‘crowd’	Lith. <i>talpa</i>	Inherited Balto-Slavic
29	* <i>tōlstō</i>	‘thick’	Lith. <i>tulžti</i>	Inherited Balto-Slavic
30	* <i>blōxa</i>	‘flea’	* <i>b^hlúseh₂</i>	Inherited PIE (non-syll.)
31	* <i>kōrjv</i>	‘root, shrub’	* <i>ker-</i> (full grade)	Full-grade
32	* <i>kōrnō</i>	‘cropped’	*(<i>s</i>) <i>ker-no-</i>	Full-grade
33	* <i>kōrvō</i>	‘ox’	* <i>ker-wo-</i>	Full-grade + * <i>w</i>
34	* <i>gōrstv</i>	‘handful’	* <i>g^her-st-</i>	Full-grade
35	* <i>sōlati</i>	‘to send’	* <i>sel-</i>	Full-grade
<i>Conditioned: PIE syllabic resonant + labial/round (n = 7; see §3)</i>				
36	* <i>gōrdlo</i>	‘throat’	* <i>g^wṛh₃-d^hlom</i>	Labiovelar
37	* <i>gōrnō</i>	‘furnace’	* <i>g^wṛ₂-nó-s</i>	Labiovelar
38	* <i>bōrzō</i>	‘fast’	* <i>b^hṛg^hús</i>	<i>u</i> -stem
39	* <i>drōvo</i>	‘wood’	* <i>dṛ₂-w-óm</i>	Labial * <i>w</i>
40	* <i>tōrkō</i>	‘run’	* <i>tṛk^w-os</i>	Labiovelar
41	* <i>mōlva</i>	‘speech’	* <i>m₂lwH-eh₂</i>	Labial * <i>w</i> + * <i>H</i>
42	* <i>tōlkō</i>	‘interpretation’	* <i>tlk^w-</i>	Labiovelar
<i>Uncertain/substrate/expressive (n = 15)</i>				
43	* <i>tōrgō</i>	‘market’	No PIE etymology	Substrate
44	* <i>gōrbō</i>	‘hump’	* <i>ger-</i> “to bend”?	Full-grade?
45	* <i>bōrkō</i>	‘moustache’	Uncertain	Expressive?
46	* <i>bōrō</i>	‘millet’	Unknown	Wanderwort?
47	* <i>kōlzati</i>	‘to slide’	Expressive	Non-inherited
48	* <i>skōrbv</i>	‘sorrow’	* <i>skreb^h-?</i>	Expressive?
49	* <i>kōlpv</i>	‘spoonbill’	Unknown	Substrate?
50	* <i>sōrkati</i>	‘to sip’	Onomatopoetic	Expressive
51	* <i>xōrtō</i>	‘greyhound’	*(<i>s</i>) <i>keur-</i> or * <i>ser-</i> ?	Uncertain
52	* <i>gōrčv</i>	‘cramp’	* <i>ger-</i> “to bend”	Expressive?
53	* <i>kōrčv</i>	‘stump’	*(<i>s</i>) <i>ker-</i> “to cut”	Expressive?
54	* <i>stōbōrō</i>	‘stalk’	*(<i>s</i>) <i>tewb^h-</i>	Uncertain
55	* <i>pōrxō</i>	‘scurf’	Expressive doublet	Expressive
56	* <i>mōrgati</i>	‘to blink’	Expressive doublet	Expressive
57	* <i>smōrkō</i>	‘mucus’	Onomatopoetic	Expressive

Not a single back-ger word traces to an unconditioned PIE syllabic resonant. The seven words in the “conditioned” row (**gʷrdlo*, **gʷrnʷ*, **bʷrʷzʷ*, **drʷvo*, **tʷrkʷ*, **mʷlva*, **tʷlkʷ*) do contain PIE syllabic resonants, but their back jer is explained by labial/round conditioning (§3). Among the remaining 50, not one derives from a PIE syllabic resonant.

Two classification decisions merit comment. First, **gʷrdʷ* ‘proud’ (#9) involves a labiovelar **gʷ*, like conditioned **gʷrdlo* (#36), but is classified under “laryngeal sequences” because its PIE form **gʷrH-dʰh₁o-* does not contain a syllabic resonant: the laryngeal **H*, not the liquid, provides the nuclear element. Compare **gʷrh₃-dʰlom* → **gʷrdlo*, where **r* is explicitly syllabic. Second, **sʷlnʷce* ‘sun’ (#21) derives ultimately from PIE **séh₂wl*, which contains syllabic **l*, but the Slavic form goes through a suffixed derivative **suh₂l-n-iko-* where the vowel **u* (from **h₂w* resolution) precedes **l*, leaving **l* non-syllabic. This differs from **drʷvo* (#39), where **r* in **dʷr-w-óm* is unambiguously syllabic before **w*.

Nine entries (#7, #31–35, #55–57) are added in the present revision following a reanalysis of Kuryłowicz’s (1956: 229–234) **uR* list (§6.4). Four entries previously listed in the “Uncertain/substrate/expressive” subgroup have also been reclassified on closer examination of cognates: **tʷlpa* and **tʷlstʷ* have solid Lithuanian/Latvian cognates (Balto-Slavic); **blʷxa* continues PIE **bʰlúseh₂* (Lith. *blusa*, Skt. *plúṣi*, Gk. *psúlla*; the **u*-vocalism is regular, not expressive); **tʷlkʷ* from zero-grade **tʷlkʷ-* (cf. Skt. *tarka*, Lat. *loquor*) belongs in the conditioned row by labiovelar **kʷ*. None of these reclassifications affects the diagnostic: all four had back jer with no syllabic-resonant source, and each now has an explicit non-syllabic-resonant etymology.

2.5 The 2×2 table

Table 5 summarizes the corpus. The seven conditioned back-ger words (PIE syllabic resonant + labial/round environment) are shown separately to maintain transparency; the four front-ger loanwords and seven uncertain front-ger words are also counted separately.

Table 5: Jer color vs. etymological source in PS liquid contexts ($n = 128$).

	PIE <i>*r</i> / <i>*l</i>	Conditioned <i>*r</i> / <i>*l</i> ^a	Non-syllabic	Loan./unc.	Total
ʙ + liquid	60	0	0	11	71
ɮ + liquid	0	7	50	—	57
Total	60	7	50	11	128

^aPIE syllabic resonant with adjacent labial/round element (§3).

The diagonal dominance is complete for unconditioned inherited words: every front-ger word traces to a PIE syllabic resonant ($n = 60$), and every back-ger word without labial conditioning traces to a non-syllabic source ($n = 50$). The seven conditioned

words represent a regular, phonologically transparent process (§3), not exceptions to the diagnostic.

A formal significance test on this table is uninformative: the corpus is hand-classified by the very hypothesis being tested, and statistical significance does not measure independence of classifier and outcome. The more meaningful question is robustness under adversarial reclassification of the borderline cases. Two stress tests are reported here. (i) *Adversarial uncertain reclassification*. If all 15 entries in the back-jer “Uncertain/substrate/expressive” group (Table 4) were reassigned to “Inherited non-syllabic-resonant”—i.e., assumed to be from full-grade or non-syllabic sources—the diagonal remains complete: 60 / 0 / 0 / 65. If instead they were all reassigned to “Conditioned syllabic resonant” (the maximally adverse direction), the diagonal still holds, but the conditioning set would need to grow to absorb them; the falsification clauses (§6.6) make this move costly. (ii) *Removal of the conditioned cases*. Dropping the seven conditioned back-jer entries entirely yields a 60 vs. 50 table with the same diagonal pattern. The empirical claim is therefore not contingent on the conditioning analysis: even without §3, no back-jer corpus entry derives from an *unconditioned* PIE syllabic resonant.

2.6 Finnish confirmation

Finnish borrowings from Proto-Slavic preserve jer vowel quality as full vowels, confirming that the distinction was phonetically real (Kallio 2006). The available Finnish data in liquid contexts are limited, but all four attested cases match the diagnostic:

Table 6: Finnish loanwords from Proto-Slavic confirm jer color in liquid contexts.

Finnish	PS source	Jer	Finnish vowel	Prediction
<i>turku</i> ‘market’	* <i>tǫrgǫ</i>	Ъ	/u/	✓ back
<i>pirta</i> ‘reed’	* <i>bǫrdo</i>	Ь	/i/	✓ front
<i>sirppi</i> ‘sickle’	* <i>sǫrpǫ</i>	Ь	/i/	✓ front
<i>virsta</i> ‘verst’	* <i>vǫrsta</i>	Ь	/i/	✓ front

The Finnish data establish that at the time of early Slavic–Finnic contact, the back jer was perceived with [u]-like coloring and the front jer with [i]-like coloring in liquid contexts. The distinction was not merely orthographic: it was acoustically detectable to non-Slavic borrowers and was systematically resolved into Finnish phonotactics.

The phonetic status of the source segment, however, is not resolved by the Finnish data alone. Two readings are compatible with the borrowing pattern. (i) The standard reading: the source segments were already full vowels of [i]-like and [u]-like quality, faithfully copied into Finnish. (ii) The alternative reading (see §3.5): the source segments were still colored syllabic resonants (a syllabic *r* with [i]-like coloring vs. [u]-like coloring), which Finnish—a language that does not permit syllabic resonants in its native

phonotactics—resolved by inserting a vowel reflecting the perceived coloring. Under reading (ii), *pirta* and *turku* record the same color distinction but via Finnic phonotactic resolution of a syllabic source, not via direct vocalic copy. Both readings agree that the color distinction was real and phonetically perceptible at the time of borrowing; they differ on the segmental category of the source. The corpus evidence in §3.5 favors reading (ii), but neither reading is excluded by the Finnish data themselves.

3 Conditioning environments

Seven Proto-Slavic words contain PIE syllabic resonants yet show back jer: **gǝrdlo* ‘throat’, **gǝrnǝ* ‘furnace’, **bǝrzǝ* ‘fast’, **drǝvo* ‘wood’, **tǝrkǝ* ‘run’, **mǝlva* ‘speech’, and **tǝlkǝ* ‘interpretation’. All seven are explained by a single principle: a labial or round segment in the immediate phonological neighborhood of the syllabic resonant conditions **uR* (back) rather than the default **iR* (front).

3.1 Labiovelars (Vaillant 1950)

PIE **g^wǝh₃-d^hlom* → PS **gǝrdlo* ‘throat’. The labiovelar **g^w* provides the rounding environment. Lithuanian *gurklỹs* ‘throat’ confirms PBS **u* vocalism. The same environment explains **gǝrnǝ* ‘furnace’ from **g^wǝh₃-nó-s* (labiovelar **g^wh*), **tǝrkǝ* ‘run’ from **tǝrk^w-os* (following labiovelar **k^w*), and **tǝlkǝ* ‘interpretation’ from zero-grade **tǝlk^w-* (full grade **telk^w-* ‘to push, urge’, with Sanskrit *tarka* ‘conjecture’ as o-grade, Latin *loquor* ‘to speak’, and OIr *do·tluchethar* ‘requests’); back jer is regular from the labiovelar **k^w* following the syllabic resonant.

The converse also holds: when a labiovelar was palatalized before a front vowel, the rounding was overridden. PIE **k^wǝmis* → PS **čǝrvǝ* ‘worm’, with front jer despite the labiovelar. Kuryłowicz (1956) observed that palatovelars never produce **uR*; our data extend this to palatalized labiovelars.

3.2 The *u*-stem

PIE **b^hǝǵ^hús* ‘short’ → PS **bǝrzǝ* ‘fast’. The stem-final **-u-* conditions back jer, just as a preceding labiovelar does. Greek *βραχύς* ‘short’ and Latin *brevīs* < **mǝǵ^hwis* confirm the PIE reconstruction.

3.3 Labial **w*

PIE **dǝ-w-óm* (thematic derivative of **dóru* ‘tree’) → PS **drǝvo* ‘wood’. The labial semivowel **w* immediately following the syllabic resonant provides the rounding environment. Compare **dǝrǵati* ‘to hold’ from **d^hǝǵ^h-* (no labial → front jer).

3.4 A feature-based generalization

All three environments share a common articulatory feature: lip rounding on a tautosyllabic segment adjacent to the syllabic resonant. The directionality is not uniform—labiovelars precede the resonant, *u*-stems and labial **w* follow it—so the generalization is not a single phonological rule applied in a fixed structural position. Rather, the diagnostic appeals to a featural condition: a round element anywhere in the immediate tautosyllabic neighborhood of the syllabic resonant shifts the PBS reflex from default **iR* to **uR*. This kind of feature-based, position-flexible spread of rounding is independently attested in the typological literature on vowel-consonant coarticulation, but the present paper makes no claim to derive it from a single ordered rule.

Table 7: Unified labial/round conditioning of PBS syllabic resonant reflexes.

Environment	Mechanism	Example	Jer
Labiovelar <i>*g^w / *k^w</i>	Adjacent round C	<i>*gǝrdlo + 2</i>	ᵝ
Palatalized labiovelar	Rounding overridden	<i>*čǝrvǝ</i>	ᵝ
<i>u</i> -stem (<i>*-u-</i>)	Following round V	<i>*bǝrzǝ</i>	ᵝ
Labial <i>*w</i>	Following round glide	<i>*drǝvo, *mǝlva</i>	ᵝ
Default (no round element)	<i>*iR</i>	<i>*vǝlkǝ</i>	ᵝ

The default PBS reflex of a PIE syllabic resonant is **iR* (front jer). A labial/round element in the immediate phonological neighborhood shifts this to **uR* (back jer), unless the labial is neutralized by palatalization.

3.5 The conditioning is consonantal: implications for segmental category

The conditioning environments in §3 share a feature that is more revealing than their unified rounding component: **every conditioner is a consonant or a glide. None is a vowel.** Table 7 can be reorganized along this axis. Labiovelars (**g^w*, **k^w*, **g^w_h*) are consonants. The labial semivowel **w* in **d_ǝ-w-óm* ‘wood’ and **m_ǝlwH-eh₂* ‘speech’ is a non-syllabic glide. The *u*-stem suffix in **b^hrg^hús* ‘short’ is the syllabic allophone of the same glide: ***u* and ***w* are allophones of a single PIE segment, distributed by syllable structure (***w* prevocally, ***u* elsewhere). The palatalization that overrides labial conditioning in **čǝrvǝ* ‘worm’ is itself a consonantal feature. The set of conditioners is uniformly {consonant, glide}; vowel-quality features—height (low/mid/high), backness independent of rounding, ATR—are absent as conditioners.

This uniformity is suggestive but not demonstrative. As a general typological tendency, vowels are conditioned predominantly by adjacent vowels (vowel harmony, distance assimilation) and by syllable structure (stress, length, position relative to nuclei);

sonorant consonants, including syllabic ones, are conditioned by their consonantal neighborhood (rounding spread, palatalization, gemination, sandhi sonorization). Reduced vowels are notoriously susceptible to harmonic conditioning from full vowels in adjacent syllables. We acknowledge the typological caveat: vowels can be conditioned by adjacent consonants (umlaut, palatalization), so the absence of vowel-quality conditioning here is not in itself proof of a consonantal target. What it provides is a defeasible argument: in a corpus where reduced vowels would be especially expected to show vowel-quality conditioning effects, the complete absence of such effects is consistent with—though not by itself demonstrative of—a consonantal target. The stronger leg of the segmental-category argument is the vocalization asymmetry developed in the companion paper (Jevremović 2026); the present subsection contributes a second, weaker line of evidence consistent with the same conclusion.

Two further observations support this reading. First, the conditioning is *positionally flexible* (§3.4): the round element may precede the syllabic resonant (labiovelar) or follow it (*u*-stem, labial **w*), with no fixed structural position. This is the profile of feature spreading across a syllabic-consonant nucleus, not of an ordered phonological rule operating on a vocalic nucleus. Second, the palatalization override in **črvv* is straightforwardly explained as feature antagonism on the conditioning consonant: a palatalized labiovelar has its rounding neutralized, and the underspecified syllabic resonant accordingly receives the default front-coloring. This is the kind of underspecification analysis natural for a sonorant nucleus, harder to motivate for a reduced vowel.

If accepted, this reading reframes what an OCS scribal ѣ or ѥ in a liquid context represents. The standard interpretation is that PBS **iR*/**uR* were already vocalized sequences in late PBS, with the vowel quality being the relevant feature; the OCS jer letters then mark short reduced vowels of [i]-like and [u]-like quality, respectively. The reading developed here points to a different interpretation: the OCS jer letters in liquid contexts are notational conventions for a still-syllabic resonant, with the choice of ѣ vs. ѥ reflecting the consonantal-environment-derived coloring rather than the vocalic quality of an intervening reduced vowel. Under this reading, the vocalization of syllabic resonants in Slavic is later than commonly assumed—possibly post-PBS, into early Common Slavic or even into the individual Slavic languages.

The chronological implication should be stated with care. The diagnostic establishes that the color distinction originated from consonantal conditioning at the period when the conditioning was active; it does *not* establish when the segment ceased to be syllabic. The two endpoints are independent. The Finnish borrowings discussed in §2.6 are equally compatible with the segment still being a colored syllabic resonant at the time of Slavic–Finnic contact (Finnic does not permit syllabic resonants and would insert a perceived coloring vowel) or with the segment having just vocalized while retaining the inherited color. The diagnostic thus constrains the *origin* of the color distinction more

strongly than its endpoint.

Two consequences follow. First, the argument here converges with the independently-developed Havlík’s law asymmetry argument in the companion paper (Jevremović 2026): there, Czech strong-position jers vocalize to /e/ in 100% of non-liquid cases but 0% of liquid-adjacent cases, suggesting that liquid-context jers were not real reduced vowels subject to ordinary Havlík vocalization. Two independent diagnostics—vocalization behavior (Havlík asymmetry) and phonological conditioning (this section)—both fail under the “real reduced vowel” hypothesis and both predict under the “notational convention for a colored syllabic consonant” hypothesis. The convergence is not a coincidence; it follows from the segmental category of the underlying object (§6.1). Second, the conditioning environments of §3 are no longer an *ad hoc* cleanup operation to save the diagnostic against counterexamples; they are positive predictions of the syllabic-consonant hypothesis, of exactly the form that hypothesis would generate. The **iR/ *uR* distribution is what it is because the segment being conditioned was a syllabic consonant.

3.6 Stability and the chronological argument

The categorical precision of the OCS jer-color distribution in liquid contexts is striking when set against the chronology of Slavic literacy. Early Gothic loans such as **dǫlgǫ* ‘debt’ (from Gothic *dulgs*) entered Slavic in the period of intense Gothic–Slavic contact roughly four to seven centuries before the canonical OCS manuscripts (Codex Marianus, Codex Zographensis, Codex Suprasliensis, 10th–11th c.). The OCS scribes distinguish **dǫlgǫ* ‘debt’ from **dǫlgǫ* ‘long’ consistently and extend the same consistency to the entire inherited corpus of jer-plus-liquid words. Since native-speaker pronunciation is what the scribes were recording, the distinction must have been phonemically stable in the spoken language across the 600-year interval between the Gothic-loan period and OCS attestation. The interesting question is not how Cyril “knew” the distinction—he was a literate compiler recording what speakers said—but why the distinction was phonemically stable in oral transmission for so long.

This stability is a structural fact about Common Slavic, not a feat of unwritten memory. Under the “real reduced vowel” hypothesis it is puzzling: reduced vowels are typologically susceptible to drift, merger, and harmonic effects across centuries. A categorical [u]-quality vs. [i]-quality distinction in a reduced-vowel position, maintained for six hundred years of oral transmission, would require a degree of phonological conservatism that is hard to motivate from typological parallels. Under the syllabic-consonant reading developed in §3.5, the stability follows naturally: the color distinction is anchored in the *consonantal environment*, which is itself structurally stable, and the segment being colored is a syllabic resonant whose realization is determined by its phonotactic context rather than by a fragile vocalic target. The phonotactic structure does the work that

“oral tradition” would otherwise have to do.

The same reading dissolves the apparent paradox posed by the four front-jer late-Christian loanwords (**lvv̥*, **kr̥vst̥*, **c̥vrky*, **lvst̥*; §2.3). At first glance it is surprising that early Germanic loans entering Slavic centuries before literacy follow the inherited **uR* pattern with categorical precision (**d̥zlg̥*, **x̥ɔlm̥*, **p̥ɔlk̥*), while late Christian-era loans entering Slavic at the very moment of literacy diverge by taking front jer. But the divergence is not from the synchronic system—it is the synchronic system. By the 9th century, the inherited **iR*+liquid pattern was numerically dominant and phonotactically productive (Table 3, $n = 60$ inherited entries against the smaller **uR* set). New loans with front-vowel sources (Latin *leō*, *Christus*; Greek *kyriakón*; Gothic *lists*) were integrated into the language by the same speakers who pronounced the inherited vocabulary, and they entered the productive front-coloring template. From the speaker’s perspective there is no anomaly: the same rule that assigns front coloring to **v̥lk̥* assigns it to **lv̥v̥*. Cyril records both faithfully because his informants pronounced both consistently.

The chronology of Slavic literacy thus does not pose a problem for the diagnostic. On the contrary: the categorical precision of OCS jer assignment is itself evidence that the distinction was a stable phonemic feature of spoken Slavic, not a notational invention of Cyril, and the four “outlying” Christian-era loans are evidence that the system was still actively productive at the time of borrowing. A vocalic-quality reading struggles to accommodate both facts simultaneously; the syllabic-consonant reading accommodates both as predictions of the same structural account.

3.7 The etymological diagnostic as a downstream consequence

The etymological diagnostic of §2–§4 (front jer marks PIE syllabic resonants, back jer marks other sources) and the segmental-category claim of §3.5–§3.6 (the OCS jer letter in liquid contexts notates a colored syllabic consonant) are not parallel observations. They are linked by a structural mechanism: every input to the liquid-context position in Slavic is mapped onto the same phonotactic slot—a syllabic-resonant nucleus—and the input’s frontness/backness feature is preserved as coloring on that slot. The etymological correlation is a downstream consequence of this mapping.

Slavic phonotactics in the position under study admits exactly one segment type: a syllabic-resonant nucleus between two consonants (or word-edge equivalents). Material entering Slavic in this position—whether inherited PIE ***r̥*/***l̥*, a borrowed CVC sequence with a foreign vowel, or substratum material of unknown phonemic shape—is mapped to that slot. The mapping is segmentally uniform across sources; what varies is the coloring of the resulting syllabic consonant. Two channels feed the coloring:

1. *Inherited channel.* A PIE zero-grade ***r̥* or ***l̥* enters Slavic already as a syllabic

consonant. Its coloring is determined by the synchronic phonological environment at the time of mapping: default front coloring (ɪ) when no labial/round element is adjacent; back coloring (ʊ) when a labiovelar, *u*-stem, or labial **w* is adjacent (§3).

2. *Source-vocalism channel*. A loanword or substratum item with a CVC frame in the source language has its source vowel mapped onto the receiving syllabic-resonant slot. The vowel’s frontness/backness feature transfers to the slot as coloring: a back-vowel source (Germanic **u*) yields back coloring (ʊ); a front-vowel source (Latin **i*, **e*; Greek **i*) yields front coloring (ɪ).

Both channels produce the same output type: a colored syllabic consonant. The OCS scribe records only the output, with the same letter (ʊ or ɪ) regardless of which channel produced it. The etymologist working backward from the output infers the channel from supporting evidence (cognate distribution, semantic field, phonological environment), and from the channel infers the etymological source.

The etymological correlation between back jer and “non-inherited PIE syllabic resonant” holds empirically because the historical distribution of inputs into Slavic is asymmetric in a useful way: nearly all inherited material that reaches this slot from PIE is from a zero-grade ***j/* root; nearly all Germanic-era loans (3rd–6th c.) had **u*-source vocalism, because Proto-Germanic had already generalized PIE syllabic resonants to **u+R* (Gothic *dulgs* from PIE ***dlǵʰos*); the four canonical late-Christian loans (Latin *leō*, *Christus*, Greek *kyriakón*, Gothic *lists*) had front-vowel sources; and substrate items, by hypothesis, are not inherited PIE syllabic resonants. The diagnostic is therefore not a direct test of “loan vs. inherited” but of “which channel and which coloring,” from which etymological status is inferred given the historical input distribution.

This framing resolves the residue of apparent exceptions in a uniform way. Full-grade PIE roots (**kər̥kə* from **(s)ker-*; **stəlbə* from **stel-bʰ-*; **kər̥jə* from **ker-*) yield back jer not because they are loanwords but because their PIE source is not a zero-grade syllabic resonant—they go through the source-vocalism channel just as loans do, with the full-grade ***e* or ***o* reanalyzed as coloring on the slot. The labial-conditioned items (**gərdlo*, **gərnə*, **tərkə*, **təlkə*, **drəvo*, **məlva*, **bər̥zə*) yield back jer because they entered the inherited channel but encountered a rounding conditioner in the immediate environment. The Christian-era front-jer loans (**ləvə*, **krəstə*, **cərky*, **ləstə*) yield front jer because their source-vocalism channel input was a front vowel. Substrate items (**tərgə*, **bər̥ə*, **kəlpə*) yield back jer because their source-vocalism input, whatever its precise phonemic shape, mapped that way. All categories receive the same structural treatment; the differences among them are differences in input channel, not differences in output type.

The diagnostic is thus a derivative of a more basic structural account, not a coin-

cidence between two independent observations. The structural account says: there is one slot, two channels, and a coloring feature preserved through the mapping. The etymological diagnostic exploits this account, together with the historical distribution of inputs, to classify Slavic vocabulary by source. The classification is reliable because the structural mapping is uniform and the historical input distribution is asymmetric—not because back jer and “non-inherited” are mystically linked.

4 Reclassifications and predictions

4.1 **v̥rba* ‘willow’

**v̥rba* is not included in Derksen’s (2008) *Etymological Dictionary of the Slavic Inherited Lexicon*; Matasović (2024) discusses it as either substratum or Proto-Indo-European in origin. The case for its inclusion in the present corpus rests on comparative evidence external to Slavic. Lithuanian *virbas* ‘rod, twig’ and Latvian *virba* ‘rod, spear’ show **i* vocalism, the regular Baltic reflex of PIE syllabic **ṛ*; Greek *ῥαπῆξ* ‘branch, twig’ shows the Greek treatment of the same syllabic resonant before a labial obstruent; Latin *verbēna* ‘sacred branches’ preserves the full grade of the same root. The reconstruction PIE **w̥r̥bʰ-eh₂* (zero-grade of **werbʰ-*) is supported by all three external witnesses, and the Slavic front jer is the predicted reflex.

A methodological caveat is in order. Reclassifying **v̥rba* on the basis of its Slavic front jer alone would be circular: the diagnostic would be selecting the reconstruction that fits it. The argument is licit here only because the Baltic and Greek cognates supply *independent* evidence for the zero-grade, converting it from a hypothesis driven by Slavic into one supported by external comparanda. Matasović’s (2024) substratum hypothesis would also be consistent with the front jer if the loan postdates the loss of syllabic resonants in Slavic; the diagnostic does not adjudicate between PIE-inherited and substratum readings, only between syllabic-resonant and non-syllabic-resonant sources at the relevant period.

4.2 **b̥rdo* ‘hill’

**b̥rdo* is likewise not included in Derksen (2008). The reconstruction PIE **b̥ṛh₃-dʰo-* (zero-grade of **b̥ṛh₃-* ‘to hit with a sharp instrument’ + nominal suffix **-dʰo-*) is supported by two independent lines of evidence. First, Latvian *birds* ‘loom, weaver’s reed’ is a regular cognate with semantic specialization; the Russian doublet бердо ‘weaver’s reed’ (vs. S. Slav. *brdo* ‘hill’) preserves the same specialization within Slavic. Second, Serbo-Croatian *b̆rdo* carries fixed initial acute (accent paradigm *a* in the standard Stang–Illič-Svityč classification; Stang 1957, Illič-Svityč 1963). Under Hirt’s law (1895; modern formulation in Illič-Svityč 1963), an acute accent on a non-final syllable of a Balto-Slavic

mobile paradigm reflects retraction triggered by a laryngeal in the immediately following syllable. The acute on *b̃rdo* is therefore independent evidence for the laryngeal **h₃* in the underlying zero-grade form **b^h_rh₃-d^ho-*, with retraction onto the syllabic resonant **r* producing the attested accent class.

The Slavic front jer is then the regular reflex of syllabic **r* in liquid context. Compare **g̃rdlo* ‘throat’ from **g^w_rh₃-d^hlom*: same laryngeal **h₃* and same suffix **-d^h-*, but **g̃rdlo* has the labiovelar **g^w* (which conditions back jer; §3) while **b̃rdo* has a plain labial **b^h* (which does not). The minimal contrast between the two confirms the conditioning analysis.

The same caveat as for **ṽrba* applies: the zero-grade reading is supported by the Baltic cognate and the accentological evidence, not by the Slavic front jer alone.

4.3 **m̃oldni* ‘lightning’: reclassified

OES м̃лнии shows back jer, and PIE **m̃ld^h-nih₂* (from **meld^h-* ‘lightning’) has syllabic **l̃*. This appears to be a counter-example. However, Latvian *milna* ‘hammer of Perkons’ confirms PBS **m̃ildnī* with **i* (front). All other PS **m-* + syllabic resonant words have front jer: **m̃rtṽz̃*, **m̃rk̃ñq̃ti*, **m̃r̃va*, **m̃r̃zñq̃ti*. Derksen (2008) lists both PBS **m̃ildnī* and **m̃uldnī* as possible. Wiktionary records both **m̃ol̃ñvj̃i* and **m̃ol̃ñvj̃i*.

The OES back jer is best explained as secondary East Slavic labialization—the same process that turned **ṽlk̃z̃* into Russian волк (with /o/ from ъ, not /e/ from ь). If it applied to **ṽlk̃z̃* → волк, it would apply equally to **m̃oldni* → м̃лнии. The original PS form was **m̃oldni* (front jer), and this word is not a counter-example.

A South Slavic OCS manuscript with м̃лнии (front jer) would be conclusive. The comparative argument—Latvian, all other **m-* words, Derksen’s dual listing—is strong but not decisive without direct manuscript evidence.

4.4 Predictions for disputed etymologies

The diagnostic generates testable predictions for six PS words of uncertain or disputed origin. In each case, the back jer ъ predicts that the word is **not** from a PIE syllabic resonant.

Table 8: Diagnostic predictions for disputed etymologies.

PS form	Meaning	Jer	Prediction
<i>*xǝrtǝ</i>	‘greyhound’	ɤ	Not from PIE syllabic resonant. Excludes simple zero-grade <i>*k_ṛt-</i> . Points to laryngeal root <i>*kerH-t-</i> or non-IE substrate.
<i>*gǝrbǝ</i>	‘hump’	ɤ	If from <i>*ger-</i> “to bend,” then full-grade or <i>o</i> -grade, not zero-grade <i>*g_ṛb-</i> .
<i>*bǝrkǝ</i>	‘moustache’	ɤ	Consistent with Germanic loanword or expressive.
<i>*skǝrbǝ</i>	‘sorrow’	ɤ	If from <i>*skreb^h-</i> , then full-grade, not <i>*sk_ṛb^h-</i> . OCS скръбь confirms early back jer.
<i>*bǝrǝ</i>	‘millet’	ɤ	No PIE etymology; back jer predicts wanderwort or substrate.
<i>*kǝlpǝ</i>	‘spoonbill’	ɤ	No PIE etymology; predicts substrate.

These predictions are falsifiable: any of these words could in principle be shown to derive from a PIE syllabic resonant. If such a derivation were established, it would constitute a counter-example to the diagnostic.

4.5 A constraint on comparative reconstructions: **tǝrgati* vs. Sanskrit *stṛhāti*

The diagnostic also functions as a constraint on comparative reconstructions outside Slavic. Vasmer (s.v. топгати) tentatively compares Slavic **tǝrgati* ‘to tear’ with Sanskrit *stṛhāti* ‘to crush, strike down’, with further doubtful links to Latin *trahō* and Gothic *þairkō*. The proposal looks attractive on the surface: the semantics align (‘tear / crush / strike’), the consonantal skeleton matches (*s-t-r-V-g^(h)-*), and the liquid-vowel-liquid shape is shared. To anyone working without an explicit jer constraint, this looks like a plausible PIE cognate set continuing zero-grade **(s)t_ṛg^h-*.

Sanskrit *stṛhāti* is a regular Class VI thematic present (the *tudāti*-type): zero-grade root + accented *-á-* thematic vowel + primary endings, from PIE **st_ṛg^h-é-ti*. The /ṛ/ is the standard Sanskrit reflex of PIE **ṛ*, preserved without epenthesis. Nothing is anomalous on the Indic side. If Slavic **tǝrgati* continued the same PIE source, however,

the diagnostic predicts front jer Ъ (§2.3): the root $*(s)trg^h-$ contains plain $*t$, $*r$, $*g^h$ with no labial, no u -stem, no labial $*w$, and no other conditioning environment from §3. The predicted Slavic outcome is $**strǫgati$ / $*tǫrgati$; the attested form is $*tǫrgati$.

The mismatch is categorical: the wrong jer color in a position the diagnostic constrains most tightly. Two readings of the resemblance remain. (i) The Sanskrit-Slavic comparison is coincidental: Class VI zero-grade thematic presents are common enough in Indic that an $s-t-r-g^h$ shape can arise without any contact with Slavic, and $*tǫrgati$ is best read as substrate or substrate-derived, paired with the substrate noun $*tǫrgǫ$ ‘market’ (Table 4). (ii) The two words continue different ablaut grades of a PIE $*(s)terg^h-$, with Indic continuing the zero-grade and Slavic continuing the o-grade $*(s)torg^h-$; but in that case Slavic would regularly show $**o \rightarrow *a$ or $*o$ (cf. $*torgnǫti$ as a hypothetical o-grade), not a back jer in liquid context, so reading (ii) still requires a positive explanation for the attested Ъ-vocalism that none of the conditioning environments of §3 supplies. Either way, the most natural cognate hypothesis (shared continuation of PIE $*(s)trg^h-$) is excluded.

The case illustrates a positive use of the diagnostic that goes beyond classifying back-jer words within Slavic. It functions as a filter on comparative reconstructions: a proposed cognate set linking Slavic $CǫRC-$ to a non-Slavic syllabic-resonant reflex is automatically suspect, because regular Slavic phonology should give $CǫRC-$ in that configuration. The Vasmer comparison passes every other check (semantics, consonants, thematic-class regularity in Indic, geographic plausibility) and fails only on this one diagnostic—exactly the profile of an accidental resemblance the diagnostic is designed to catch. The same logic excludes the other two proposals for $*tǫrgati$ (Chernykh’s $*terg^h-$, Derksen’s link to Latin *tergō*): each predicts front jer under §2.3.

4.6 Loanword chronology

The loanword pattern in the corpus stratifies chronologically:

Table 9: Chronological stratification of loanwords by jer color.

Period	Pattern	Examples	Jer
Early (pre-Christian)	Gmc $*u \rightarrow$ PS Ъ	$*dǫlgǫ$, $*xǫlmǫ$, $*pǫlkǫ$	Ъ
Late (Christian era)	Adapted to Ъ+liquid template	$*lǫvǫ$, $*krǫstǫ$, $*cǫrky$	Ь

Early Germanic loans preserved source-language $*u$ as back jer; later Christian-era loans were assimilated to the productive $ǫ$ +liquid template established by inherited syllabic-resonant vocabulary (§2.3), regardless of source vocalism. The Germanic-loanword stratum in Proto-Slavic has been treated systematically by Pronk-Tiethoff (2013), whose chronological layering broadly agrees with the picture used here: pre-Christian-era loans (e.g., $*xǫlmǫ$, $*pǫlkǫ$, $*dǫlgǫ$) belong to an early stratum with Ger-

manic **u*-vocalism preserved, while later borrowings show different adaptation patterns. The broader stability argument is developed in §3.6; this subsection treats the chronological pattern as a synchronic loanword-adaptation phenomenon.

Two factors contribute to the late-period pattern, and they should be distinguished. (i) *Source-language vocalism*: *Christus* has Latin /i/, *kyriakón* has Greek /i/, and *leō* (via Germanic) has /e/ or /i/-colored vocalism. All four late loans (**lvvθ*, **krvstθ*, **cbrky*, **lbstb*) had front-vowel sources, and a straightforward sound-substitution account would predict front jer in PS regardless of any template. (ii) *Phonotactic template*: by the late PS period, the inherited *v+liquid* pattern (**vθlkθ*, **pvrstθ*, **čvrnθ*, etc.) was numerically dominant in the language’s liquid-cluster vocabulary, plausibly making *ʌ+liquid* the preferred adaptation pattern for any incoming word with a roughly compatible source vowel. These two factors are difficult to disentangle empirically: there is no late loanword in our corpus with a back-vowel source that would test whether the template overrides the source vocalism. The earlier Germanic loans (**dθlgθ*, **xθlmθ*, **pθlkθ*) had source /u/ and emerged with back jer, consistent with either account. The honest reading is therefore that source vocalism does most of the explanatory work for the late-period pattern, with the productive template a plausible but not independently demonstrated reinforcement. The diagnostic’s strength does not depend on resolving this question.

5 Comparison with Winter’s law

The jer diagnostic is not the first etymological fossil in Balto-Slavic. Winter’s law (Winter 1978; Kortlandt 1988) describes vowel lengthening before PIE voiced (non-aspirated) stops in Baltic and Slavic. Like the jer diagnostic, Winter’s law encodes historical information about the PIE source in a synchronic phonological property.

Table 10: The jer diagnostic compared with Winter’s law as etymological fossils in Balto-Slavic.

Diagnostic	PIE input	BS output	Accuracy
Winter’s law	Voiced stop	Long vowel	~90%
Jer diagnostic (positive)	Syllabic <i>*r̥/ *l̥</i>	Front jer ʌ	~100% (inherited)
Jer diagnostic (negative)	Non-syllabic source	Back jer ɔ	100%

The two diagnostics operate in orthogonal domains and do not interact. Winter’s law conditions vowel *length* (lengthening before voiced unaspirated stops); the jer diagnostic discriminates the *coloring* (ʌ vs. ɔ) of a syllabic-nucleus position in liquid contexts. The PIE environments are likewise disjoint: Winter targets voiced unaspirated stops (***b*, ***d*, ***g*, ***ǵ*, ***g^w*), while the jer color is fixed by the presence vs. absence of a PIE syllabic resonant in the nucleus, irrespective of the following obstruent’s voicing. A corpus entry such as **sʌrdʌce* ‘heart’ from **k̑rd-* has PIE ***d* (a Winter environment)

and inherited syllabic **r* (front jer) simultaneously, but the two facts answer different questions about the word: Winter predicts lengthening of the nuclear vowel where one is present; the jer diagnostic predicts the color of the syllabic-nucleus position in a liquid context. Neither subsumes the other.

A deeper orthogonality holds under the segmental-category reading developed in §3.5. Winter’s law operates on PIE vowels and produces length distinctions in PBS vowels—it is fundamentally a vocalic process, targeting vowel nuclei. If the “jer” in liquid contexts is not a vowel at all but a colored syllabic consonant, then Winter’s law and the jer diagnostic operate on *categorically different segment types*: one on vowels, the other on syllabic consonants. They cannot interact because they are working on different categories of segment, not merely on disjoint phonological environments within the same segmental category. The non-interaction is thus expected at a more basic level than was suggested by their environmental disjointness alone.

Winter’s law has known exceptions (analogical levelings, loanwords that mimic the pattern). The jer diagnostic’s negative direction has none: no back-jer-plus-liquid word derives from a PIE syllabic resonant. This absolute negative is the diagnostic’s greatest strength.

6 Discussion

6.1 Relationship to the Havlík’s law asymmetry

The jer diagnostic is logically independent of, but mechanistically connected to, the Havlík’s law asymmetry described in a companion preprint (Jevremović 2026). The companion paper documents that Czech strong-position jers vocalize to /e/ in 100% of non-liquid cases but in 0% of liquid-adjacent cases. That asymmetry is the vocalization-behavior analogue of the conditioning-pattern argument developed in §3.5.

The mechanism that unifies the two results can be stated explicitly. The “real reduced vowel” hypothesis for the liquid-context jers predicts two things: (a) such jers should vocalize like other jers under Havlík’s law in strong positions, and (b) such jers should show some vowel-quality conditioning (harmony, height assimilation, or similar). Both predictions fail in the data: (a) is the empirical observation of the companion paper, (b) is the observation of §3.5. The “notational convention for a colored syllabic consonant” hypothesis, by contrast, predicts both failures: a syllabic consonant is not subject to vowel-targeting rules like Havlík vocalization, and it is conditioned by its consonantal neighborhood rather than by adjacent vowels. The convergence of two independent diagnostics on the same revisionist conclusion is therefore not coincidence: it follows directly from the segmental category of the underlying object.

The present paper’s etymological diagnostic does not depend on the phonological-

category claim to be useful: even if the segmental status of liquid-context jers remained an open question, the synchronic distribution documented in §2 would still be valid as an etymological classification tool. What the connection to Havlík does is convert a purely synchronic-distributional finding into one part of a broader argument about Common Slavic historical phonology.

6.2 Non-circularity

A potential objection is that the diagnostic is circular: if PIE etymologists reconstructed syllabic $*r/$ $*l$ partly on the basis of Slavic front jer, the diagnostic merely recovers its own input assumptions. This objection does not withstand scrutiny. PIE syllabic resonant reconstructions rest on cross-branch cognates—Greek, Latin, Sanskrit, and Baltic reflexes independently establish the zero-grade forms. Lithuanian *vilkas* ‘wolf’ (with PBS $*i$ vocalism) confirms PIE $*wlk^wos$ regardless of what Slavic shows; Sanskrit *mṛtá-* ‘dead’ confirms $*mṛtwós$ independently of $*mṛtvō$. The Finnish evidence (§2.6) provides entirely external confirmation from a non-IE family: *pirta* with /i/ and *turku* with /u/ prove that jer vowel quality was phonetically real at the time of borrowing, not an artifact of etymological reconstruction.

6.3 Substrate identification

Back-jer words with no IE etymology cluster in the non-inherited vocabulary stratum: $*tǝrgǝ$ ‘market’ (Balkan areal), $*bǝrǝ$ ‘millet’ (wanderwort?), $*kǝlpǝ$ ‘spoonbill’ (substrate?), $*kǝlbasa$ ‘sausage’ (Turkic), $*bǝlvǝnǝ$ ‘idol’ (Turkic). The diagnostic can help identify substrate words systematically: if a PS word has back jer adjacent to a liquid and lacks a secure PIE etymology, it is a candidate for the non-inherited stratum.

6.4 Reanalysis of Kuryłowicz’s 1956 corpus

This section reanalyzes Kuryłowicz’s (1956: 229–234) enumerated 58 $*uR$ examples under the diagnostic developed here. Kuryłowicz organizes his list by preceding-consonant environment (after velars, dentals, labials, $*s$, $*m/$ $*n$, liquids, $*u/$ $*i$, and word-initial). Our diagnostic applies only to Slavic-internal liquid contexts ($\mathfrak{r} + l/r$); accordingly, the reanalysis restricts the corpus on three criteria:

1. **Slavic-only.** Items attested only in Baltic (Lithuanian, Latvian, Old Prussian) without a Slavic cognate, or items where the cognate is built on a different stem, fall outside the diagnostic’s scope. Kuryłowicz includes 21 such items.
2. **Liquid contexts only.** Items where the syllabic resonant is a nasal ($**\mathfrak{m}$, $**\mathfrak{n}$) rather than a liquid ($**\mathfrak{l}$, $**\mathfrak{r}$), or where the Slavic form has $\mathfrak{r} +$ nasal rather

than $\mathfrak{r}$ + liquid, fall outside the scope. Kuryłowicz includes 10 such items, mostly under his “after $*m$, $*n$ ” rubric.

3. **Genuine PIE syllabic resonant.** Items with a non-syllabic PIE source (full-grade root, laryngeal sequence, $*u$ from $**Hw$ resolution) are predicted by the diagnostic to have back jer; they confirm the diagnostic but do not test it.

After applying these filters, Kuryłowicz’s 58 $*uR$ examples reduce to a Slavic-internal liquid-context residue of 19 items. Of these, every item either (a) has a non-syllabic-resonant PIE source (predicted by the diagnostic to have back jer regardless of conditioning), or (b) sits in one of the conditioning environments of §3 (labiovelar, u -stem, labial $*w$). Table 11 presents the reanalysis.

Table 11: Reanalysis of Kuryłowicz (1956) $*uR$ examples with Slavic-internal liquid context. Item numbers follow Kuryłowicz’s enumeration.

K#	Slavic form	PIE source	Accounted for via	Tab. 4?
2	$*k\bar{o}l\bar{o}$ ‘tusk’	zero-grade $**kol-$	non-syllabic (zero-grade)	#16
7	$*k\bar{o}rj\bar{o}$ ‘root’	$**ker-$	full-grade source	new
9	$*k\bar{o}rn\bar{o}$ ‘cropped’	$**ker-no-$	full-grade source	new
10	$*k\bar{o}rpati$ ‘patch’	$**kerh_1p-$	full grade + laryngeal	#19
11	$*k\bar{o}rv\bar{o}$ ‘ox’	$**ker-wo-$	full grade + labial $*w$	new
12	$*k\bar{o}lp\bar{o}$ ‘spoonbill’	substrate	non-IE source	#43
15	$*g\bar{o}rb\bar{o}$ ‘hump’	$**ger-$	full-grade source	#38
16	$*g\bar{o}rdlo$ ‘throat’	$**g^w rh_3-dlom$	labiovelar conditioning	#30
17	$*g\bar{o}rn\bar{o}$ ‘furnace’	$**g^wh r-no-$	labiovelar conditioning	#31
18	$*g\bar{o}rst\bar{o}$ ‘handful’	$**g^h er-$	full-grade source	new
20	$*sk\bar{o}rb\bar{o}$ ‘sorrow’	$**skreb-?$	full grade / expressive	#42
28	$*st\bar{o}lb\bar{o}$ ‘post’	$**stel-b^h-$	full-grade source	#13
29	$*p\bar{o}rx\bar{o}$ / $*p\bar{o}rr\bar{o}$ ‘scurf’	onomatopoetic	expressive doublet	new
31	$*b\bar{o}rna$ ‘lip’	cf. Arm. <i>beran</i>	uncertain / non-PIE	#26
32	$*b\bar{o}rz\bar{o}$ ‘fast’	$**b^h r g^h \acute{u}s$	u -stem conditioning	#32
35	$*s\bar{o}lati$ ‘to send’	$**sel-$	full-grade source	new
41	$*m\bar{o}ldni$ ‘lightning’	$**m\bar{d}^h-$	secondary labialization	—
45	$*m\bar{o}rgati$ ‘blink’	expressive	expressive doublet	new
46	$*sm\bar{o}rk\bar{o}$ ‘snot’	onomatopoetic	expressive	new

The reanalysis yields three findings.

First, **no Kuryłowicz item with Slavic-internal liquid context derives from an unconditioned PIE syllabic resonant.** Every item is accounted for by either a

non-syllabic-resonant source (full grade, laryngeal sequence, expressive/onomatopoetic, substrate) or one of the labial/round conditioning environments of §3. The diagnostic’s 100% / 100% claim survives the test against the primary published corpus.

Second, **the diagnostic has independent support in Kuryłowicz’s own per-environment statistics** (Table 1): after palatovelars, **uR* appears in zero of 16 cases, which is the categorical limit version of our “palatalization override” observation (§3). Kuryłowicz himself drew this conclusion (1956: 229), as did Vaillant (1950: 171).

Third, **nine Kuryłowicz items were absent from earlier formulations of Table 4 and have been added in the present revision**: **kɔlějb* ‘glue’ (Germanic loan per Kiparsky 1934), **kɔrjb* ‘root, shrub’, **kɔrnɔ* ‘cropped’, **kɔrvɔ* ‘ox’, **gɔrstɔ* ‘handful’, **sɔlati* ‘to send’ (five full-grade roots), **pɔrxɔ*/**pɔrxɔ* ‘scurf’, **mɔrgati*/**mɔrgati* ‘blink’, and **smɔrkɔ* ‘mucus’ (three expressive/onomatopoetic items). Each item is accounted for by the diagnostic (full-grade source or expressive/non-inherited classification), so the central 100%/100% claim is unchanged; the corpus expansion sharpens the empirical base from 48 to 57 back-jer entries and brings the present paper’s back-jer corpus into alignment with Kuryłowicz’s published list.

The Kuryłowicz reanalysis converges with the substratum/contact hypothesis Andersen (2003: 60–62) develops on independent grounds. Andersen’s framework deserves a separate discussion: it is the most articulated published account of the PBS **iR*/**uR* distribution and arrives at a conclusion that overlaps substantially with ours.

Andersen begins from the observation that the **uR*/**iR* distribution among Balto-Slavic discrepant-dorsal lexemes (his Table 3) is the same as the distribution in lexemes where syllabic sonorants are preceded by other consonants. He takes this to mean that “all the lexemes with **uR* reflexes (or **uR* ~ **iR* doublets) reflect contact with dialects in which syllabic sonorants had a different development” (Andersen 2003: 61). Two scenarios are proposed. (i) *Areal difference*: substratum dialects from which lexemes were adopted diphthongized PIE syllabic resonants to ***uR* (as Germanic did, e.g. ***wulfaz* ‘wolf’ from **wlk^wos*), whereas inherited (pre-)Slavic–Baltic gave ***iR*. (ii) *Chronological difference*: early loans, while (pre-)Slavic–Baltic still had syllabic resonants, were diphthongized to ***iR* along with native vocabulary; later loans, when (pre-)Slavic–Baltic had already lost syllabic resonants, were adopted directly with the donor language’s ***uR*. Andersen further endorses Stang’s (1966: 79–80) semantic observation: many **uR* items cluster in expressive and onomatopoetic semantic fields—“fat”, “dumb”, “lazy”, “clumsy”, “crooked, bent”, “crippled, decrepit”, “dark”, “dirty”—reflecting a universal sound-symbolic preference for back-vowel diphthongs that may have driven **iR* → **uR* expressive replacement (Jakobson & Waugh 1979).

Andersen’s account and the present diagnostic make the same empirical prediction on Slavic liquid-context vocabulary: a Slavic word with back jer ɤ adjacent to a liquid is not directly inherited from a PIE syllabic resonant. Andersen identifies the *historical*

source of the non-inheritance (substratum intrusion, late loan, expressive innovation); the present paper identifies the *synchronic phonological signature* of that non-inheritance (the jer color itself). The two frameworks therefore converge on a single conclusion by complementary routes.

Two differences in scope and emphasis are worth noting. First, Andersen’s account is broader, covering all four PBS syllabic resonants ($*l$, $*r$, $*m$, $*n$) across Slavic and Baltic; our diagnostic restricts to liquid contexts ($*l$, $*r$) in Slavic alone. Second, Andersen does not aim to predict the back jer in any particular Slavic word, but rather to explain the dual-reflex phenomenon at the level of historical lexical stratification; the present diagnostic aims at item-level classification of Slavic vocabulary by jer color. The combination of the two yields a *both/and* reading: back-jer liquid-context words in Slavic are non-inherited (per the diagnostic), *and* they are non-inherited because of substratum contact or expressive replacement (per Andersen’s framework).

This convergence has methodological value beyond corroboration. The diagnostic was developed without reference to Andersen’s framework—it was derived from the Slavic-internal jer corpus together with the conditioning environments of §3—yet it lands on the same conclusion as Andersen reaches from his contact-linguistic premises. Where two independent lines of reasoning converge on the same empirical claim, the claim gains independent corroboration. The remaining open questions are downstream of this convergence: which specific back-jer items reflect Germanic-like substratum dialects (Andersen’s areal hypothesis), which reflect late loans (his chronological hypothesis), and which reflect expressive replacement (Stang’s hypothesis). These are open historical-linguistic questions that the synchronic diagnostic does not resolve but does help to circumscribe.

6.5 Onomastic application: $*srb̥$ vs. $*x̥rvat̥$

The two principal South Slavic ethnonyms display opposite jer colors: $*srb̥$ ‘Serb’ with front jer $̥$, and $*x̥rvat̥$ ‘Croat’ with back jer $̥$. The diagnostic predicts different etymological origins—and the evidence confirms this.

$*x̥rvat̥$ (Table 4, entry #4) is universally accepted as an Iranian loanword from Proto-Ossetian/Alanian $*xurvāt-$ ‘guardian’ (Abaev 1958–1989; Loma 2000), attested in Greek as Χοροάθος in the 3rd-century Tanais Tablets. The back jer correctly marks non-inherited status.

$*srb̥$, by contrast, has front jer $̥$ —the pattern for inherited PIE vocabulary. The mainstream etymology derives it from PIE $*ser-$ ‘to protect, guard’ (cf. Latin *servāre*, Avestan *haraiti*). An alternative etymology (Schuster-Šewc 1985) derives it from the zero-grade of PIE $*srebh-$ ‘to sip, suckle’: $*srbh-os$ ‘milk kinsman’ → ethnonym. The zero-grade $*srbh-$ is independently attested in Armenian *arb* ‘he drank’, Albanian *gjerb*

‘to sip’, and Lithuanian *surbti* ‘to suck’, and the PS verb **sɐrbati* ‘to slurp’ undeniably derives from this root.

The diagnostic does *not* adjudicate between these two etymologies. Both yield front jer: the **ser-* ‘to protect’ derivation goes through a full-grade root where the front jer would have to be explained by zero-grade derivation at some intermediate stage, while the **sɐrb^h-* ‘to suckle’ derivation explains the front jer directly through PIE syllabic **ɾ*. The Schuster-Šewc reading is therefore more economical with respect to the diagnostic—it requires no intermediate ablaut step—but it is not the only reading compatible with front jer, and reasonable scholars may prefer one or the other on independent grounds (semantic plausibility of ‘guardian’ vs. ‘milk kinsman’ as ethnonymic source). What the diagnostic does establish is that **sɐrbɔ* cannot be derived from any source incompatible with PIE syllabic **ɾ* or zero-grade ablaut—in particular, it cannot be an Iranian loanword on the model of **xɐrvatɔ*, which is the principal claim relevant to Loma (2000).

The Lusatian Sorbs (West Slavic) share the identical ethnonym, proving it is a Proto-Slavic word predating the 6th-century migrations, not a local Balkan borrowing. Ukrainian *pryserbytysja* ‘to join, attach oneself’ confirms that the root had a live appellative meaning ‘ally, kinsman’, not just a frozen ethnonymic use.

The opposite jer colors of these neighboring ethnonyms constitute independent evidence against a common Iranian origin for both (contra Loma 2000), and illustrate the diagnostic’s applicability beyond common vocabulary.

6.6 Falsification conditions

A diagnostic that absorbs every counterexample as “further conditioning” is unfalsifiable. To preempt this objection, the conditions under which the diagnostic would be refuted are stated in advance.

1. **Positive direction.** A securely reconstructed Proto-Slavic word with front jer ɐ adjacent to a liquid whose PIE source is *not* a syllabic resonant (**ɾ*/**l*), and which is not a late loanword adapted to the productive ɐ+liquid template, would refute the positive direction. The four loanwords **lbvɔ*, **krɐstɔ*, **cɐrky*, **lɐstɔ* are excluded explicitly and in advance, on independent grounds (post-Christianization date, attested non-Slavic source).
2. **Negative direction.** A securely reconstructed PS word with back jer ɔ adjacent to a liquid whose PIE source *is* a syllabic resonant, and which lacks any labial or round element in the immediate phonological neighborhood of that resonant (no labiovelar, no *u*-stem, no labial **w*, no *u*-coloring via laryngeal), would refute the negative direction. Critically, the conditioning environments of §3 are listed exhaustively; new environments cannot be added post hoc to absorb a counterex-

ample. If a back-jer word were found whose PIE syllabic-resonant source has only plain (non-labialized) consonants and a non-*u* stem, the diagnostic would fail.

3. **What does *not* refute.** Surface back jer in attested manuscripts where the comparative evidence supports an underlying front jer (the **mǫldni* case, §4.3) does not refute the diagnostic, provided an independent labialization process can be identified (here, secondary East Slavic labialization documented also for **vǫlkǝ* → Russian волк). Reviewers may reasonably disagree about whether such a reanalysis is licit in any given case; this is a limit on the diagnostic’s certainty, not on its falsifiability in principle.

The diagnostic is presented here as falsifiable in this sense: the conditioning environments are closed, not open-ended, and a single counterexample of the type in (2) would force a substantive revision. Two candidates examined in this paper (**mǫldni*, **gǫrdǝ*) have been considered and explained without expanding the conditioning set.

A complementary observation about the diagnostic’s structure is worth stating. The back-jer corpus admits classification ambiguity: some items could be analyzed either as PIE syllabic resonant + conditioning, or as full-grade native, or as loan, or as substrate, depending on which non-Slavic comparanda one weighs most. The diagnostic is robust under this ambiguity because all four channels produce the same predicted output, back jer (§3.7). The diagnostic does not adjudicate between competing back-jer-producing channels for any specific item; it makes the weaker but more general claim that no back-jer Slavic item with liquid context derives from an *unconditioned* PIE syllabic resonant in a *non-loan*, *non-substrate* context. The four channel categories are exhaustive of the attested corpus, and a refutation would require finding a back-jer item that fits none of them.

6.7 Limitations

Several limitations should be noted:

1. The diagnostic applies only to jer-plus-liquid positions (*CǝrC/CǝrC*, *CǝlC/CǝlC*). It says nothing about jer distribution in other environments.
2. The conditioning environments (labiovelars, *u*-stems, labial **w*) add complexity. The unified “labial/round” principle is transparent, but each environment must be checked before applying the diagnostic.
3. The **mǫldni* reclassification rests on comparative evidence (Latvian, all other **m*-words) rather than direct OCS manuscript attestation.
4. Some PIE etymologies are themselves uncertain. The diagnostic presupposes correct jer-type reconstruction in Proto-Slavic and correct PIE etymology.

7 Conclusion

Back jer ь adjacent to a liquid in Proto-Slavic is a reliable negative marker for PIE syllabic resonant origins. In a corpus of 128 words (57 back-jer, 71 front-jer), no PS word with back jer adjacent to a liquid derives from a PIE syllabic **r̥* or **l̥*. The seven apparent exceptions (**gǫrdlo*, **gǫrnǫ*, **bǫrznǫ*, **drǫvo*, **tǫrkǫ*, **mǫlva*, **tǫlkǫ*) are explained by a unified labial/round conditioning principle; the single manuscript-level candidate (**mǫldni*) is best analyzed as secondary East Slavic labialization.

Conversely, every inherited PS word with front jer ɨ adjacent to a liquid traces to a PIE syllabic resonant ($n = 60$, 100%). The only non-syllabic-resonant front-jer words are four late Germanic loanwords adapted to a productive phonotactic template. Two minimal pairs—nominal **dǫlgǫ*/**dǫlgǫ* and verbal **tǫrzati*/**tǫrgati*—crystallize the diagnostic, and the worked example of **tǫrgati* vs. Sanskrit *strhāti* (§4.5) shows the diagnostic functioning as a filter on comparative reconstructions.

This makes jer color a novel etymological classification tool—the first diagnostic specific to the Slavic liquid-cluster problem. It joins Winter’s law as an etymological fossil in Balto-Slavic: a synchronic phonological property that encodes diachronic information about the PIE source. Its categorical negative—no back-jer word from an unconditioned syllabic resonant—gives it a discriminating power that is, in its limited domain, unmatched.

The diagnostic is best understood as a derivative of a structural account, not as a free-standing empirical correlation. The shape of the conditioning environments (uniformly consonantal/glide; no vowel-quality conditioning; §3.5) indicates that the conditioned segment is a syllabic consonant rather than a reduced vowel. The categorical precision of OCS jer assignment across the 600-year gap between Gothic-loan-period borrowings and 9th-century literacy (§3.6) follows from the structural stability of consonantal anchoring rather than from improbable lexical memory. The etymological correlation between back jer and non-PIE-syllabic-resonant sources (§3.7) arises from the asymmetric historical distribution of inputs into a single Slavic phonotactic slot: inherited PIE **r̥*/**l̥* enter through one channel (with conditioning-dependent coloring), loanwords and substrate material enter through another (with source-vocalism-derived coloring), both produce the same output type. The etymological tool is what one obtains by working backward from coloring through channel to source.

Two diagnostics for the segmental-category claim now converge on the same conclusion: the conditioning-pattern argument developed here, and the vocalization-asymmetry argument from Havlík’s law developed in the companion paper (Jevremović 2026). Both fail under the “real reduced vowel” hypothesis and both predict under the “colored syllabic consonant” hypothesis. The convergence supports a modest but genuine revision of Common Slavic historical phonology: liquid-context OCS ь and ɨ are notational

conventions for colored syllabic resonants, not graphemes for ordinary reduced vowels. The vocalization of syllabic resonants in Slavic is correspondingly later than commonly assumed—possibly post-Proto-Slavic, into the early daughter languages—though the diagnostic constrains the origin of the color distinction more strongly than its endpoint.

References

- Abaev, V. I.** (1958–1989). *Istoriko-ètimologičeskij slovar’ osetinskogo jazyka*. 4 vols. Moscow/Leningrad: Nauka.
- Andersen, H.** (2003). Slavic and the Indo-European migrations. In H. Andersen (ed.), *Language Contacts in Prehistory: Studies in Stratigraphy*, 45–76. Current Issues in Linguistic Theory 239. Amsterdam/Philadelphia: John Benjamins. <https://doi.org/10.1075/cilt.239.05and>
- Derksen, R.** (2008). *Etymological Dictionary of the Slavic Inherited Lexicon*. Leiden: Brill.
- ESJS** = Havlová, E. et al. (1989–). *Etymologický slovník jazyka staroslověnského*. Brno: Academia / Tribun EU.
- ÈSSJa** = Trubačev, O. N. et al. (eds.) (1974–). *Ètimologičeskij slovar’ slavjanskix jazykov: Praslavjanskij leksičeskij fond*. Moscow: Nauka.
- Hirt, H.** (1895). *Der indogermanische Akzent: Ein Handbuch*. Strassburg: Trübner.
- Illič-Svityč, V. M.** (1963). *Imennaja akcentuacija v baltijskom i slavjanskom*. Moscow.
- Jakobson, R., & Waugh, L. R.** (1979). *The Sound Shape of Language*. Bloomington: Indiana University Press.
- Jevremović, M.** (2026). Havlík’s law asymmetry: Evidence against real reduced vowels in Proto-Slavic liquid contexts. LingBuzz preprint. <https://lingbuzz.net/lingbuzz/009846>
- Kallio, P.** (2006). On the earliest Slavic loanwords in Finnic. *Slavica Helsingiensia* 27, 154–166.
- Kortlandt, F.** (1988). Remarks on Winter’s law. *Dutch Contributions to the Tenth International Congress of Slavists*, 387–396.
- Kuryłowicz, J.** (1956). *L’apophonie en indo-européen*. Wrocław: Polska Akademia Nauk.
<https://archive.org/details/LAphonieEnIndoEuropeenKurylowicz>

- Loma, A.** (2000). Serbisches und kroatisches Sprachgut bei Konstantin Porphyrogenetos. *Zbornik radova Vizantološkog instituta* 38, 87–161.
- Matasović, R.** (2008). *Etymological Dictionary of Proto-Celtic*. Leiden: Brill. (p. 111 for the ordered-rule treatment of PBS syllabic-resonant reflexes.)
- Matasović, R.** (2024). Proto-Slavic forest tree names: Substratum or Proto-Indo-European origin? In *Etymological Studies in Slavic*, ed. M. Snoj. Ljubljana: ZRC SAZU.
- Mayer, H. E.** (1991). Reflexes of Indo-European syllabic resonants in Baltic, Slavic, Albanian. *Lituanus* 37(4). Online: http://www.old.lituanus.org/1991_4/91_4_05.htm
- Pronk-Tiethoff, S.** (2013). *The Germanic Loanwords in Proto-Slavic*. Leiden: Brill.
- Schuster-Šewc, H.** (1985). Zur Etymologie und Deutung des Namens Serb-/Sorb-. *Zeitschrift für Slawistik* 30, 702–707.
- Shevelov, G. Y.** (1964). *A Prehistory of Slavic: The Historical Phonology of Common Slavic*. Heidelberg: Carl Winter.
- Sławski, F. (ed.)** (1974–2001). *Słownik prastłowiański*. 8 vols. Wrocław: Ossolineum.
- Stang, C. S.** (1957). *Slavonic Accentuation*. Oslo: H. Aschehoug & Co.
- Stang, C. S.** (1966). *Vergleichende Grammatik der baltischen Sprachen*. Oslo: Universitetsforlaget.
- Vaillant, A.** (1950). *Grammaire comparée des langues slaves*, vol. I. Paris: Klincksieck.
- Vasmer, M.** (1953–1958). *Russisches etymologisches Wörterbuch*. 3 vols. Heidelberg: Carl Winter. Russian translation: Trubačev, O. N. (1964–1973). *Etimologičeskij slovar' russkogo jazyka*. Moscow: Progress.
- Winter, W.** (1978). The distribution of short and long vowels in stems of the type Lith. *ėsti* : *vėsti* : *mėsti* and OCS *jasti* : *vesti* : *mesti*. In *Recent Developments in Historical Phonology*, 253–272.